

Chapter 1 In-Class Exercise: `print()` and Numeric Expressions

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August 24, 2026

1 Printing Arithmetic in Python

1. Python can perform arithmetic directly inside the `print()` function. The multiplication operator in Python is the asterisk, `*`.

- Create a new Python file called `multi_table.py`. Begin by printing a title:

```
print("Multiplication Table")
```

- We will arrange the products in rows and columns. Begin by printing a row containing the column headings:

```
print("    ", 1, 2, 3, 4, 5)
```

- Now print the first row. Python should calculate each product rather than having you type the answers:

```
print(1, 1*1, 1*2, 1*3, 1*4, 1*5)
```

- Run the program. The numbers following the first 1 should be the products obtained by multiplying 1 by 1 through 5.

- Each entry in the table should be calculated as the product of its row number and column number. For example, the entry in row 3 and column 4 should be calculated using

```
3 * 4
```

rather than by typing 12 directly.

- Use one `print()` statement for each row of the multiplication table.
- Before writing the entire program, determine the five multiplication expressions needed to produce row 4. Sketch this `print()` statement below before beginning in Google CoLab.
- Run your program and compare its output with the example above. Check that every row and column contains five products, and that the lower-right entry is the product of 5 and 5.

2 In-Class Exercise

1. Complete the program using only simple Python statements, arithmetic expressions, and the `print()` function. Your program should produce a two-dimensional multiplication table for the first five positive integers:

Multiplication Table

	1	2	3	4	5
1	1	2	3	4	5
2	2	4	6	8	10
3	3	6	9	12	15
4	4	8	12	16	20
5	5	10	15	20	25

- Each row represents one integer from 1 through 5, and each column represents one integer from 1 through 5.